

Name: _____

Date: _____

PERCENTAGES, (COMPOUND INTEREST AND DEPRECIATION)

LO: I can calculate compound interest and depreciation using repeated percentage change.

Compound interest and depreciation

Compound interest is earned on the original amount AND on previous interest. Depreciation is when a value decreases by a fixed percentage each year.

Formula: Final value = Original $\times$ (multiplier)ⁿ, where n is the number of years.

Quick examples

●	5% increase: multiply by 1.05	compound growth multiplier
●	10% depreciation: multiply by 0.90	depreciation multiplier
●	£1000 at 3% for 2 years = 1000×1.03^2	compound formula
●	5% for 3 years = 15% total	that is simple interest

Key idea

Use the multiplier raised to the power of years. Growth: $1 + r$. Depreciation: $1 - r$.

$$A = P \times (1 + r)^n$$

P = principal, r = rate as decimal, n = years

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - compound interest

£2000 is invested at 4% compound interest for 3 years. Find the final amount.

$$\text{Multiplier} = 1 + 0.04 = 1.04$$

$$A = 2000 \times 1.04^3$$

$$A = 2000 \times 1.124864 = \text{£}2249.73$$

$$= \text{£}2249.73$$

Multiply by the growth multiplier for each year.

Example 2 - finding interest earned

£5000 at 3% compound interest for 2 years. How much interest is earned?

$$A = 5000 \times 1.03^2 = 5000 \times 1.0609$$

$$A = \text{£}5304.50$$

$$\text{Interest} = 5304.50 - 5000 = \text{£}304.50$$

$$= \text{£}304.50 \text{ interest}$$

Subtract the original to find just the interest.

Example 3 - depreciation

A car worth £12000 depreciates by 15% per year. Find its value after 4 years.

$$\text{Multiplier} = 1 - 0.15 = 0.85$$

$$V = 12000 \times 0.85^4;$$

$$V = 12000 \times 0.52200625 = \text{£}6264.08$$

$$= \text{£}6264.08$$

Depreciation uses a multiplier less than 1.

Example 4 - finding the rate

A painting increases from £500 to £605 in 2 years compound. Find the annual rate.

$$605 = 500 \times r^2, \text{ so } r^2 = 1.21$$

$$r = \sqrt{1.21} = 1.1$$

$$\text{Rate} = 10\% \text{ per year}$$

$$= 10\% \text{ per year}$$

Divide by the original, then take the nth root.

YOUR TURN – PRACTICE (deliberate practice)

1. Compound interest

Find the final amount.

- a) £1000 at 5% for 2 years
- b) £3000 at 4% for 3 years
- c) £500 at 6% for 2 years
- d) £8000 at 2% for 4 years

2. Interest earned

Find the interest earned (final amount minus principal).

- a) £2000 at 3% for 3 years
- b) £4500 at 5% for 2 years
- c) £10000 at 2.5% for 4 years
- d) £750 at 8% for 3 years

3. Depreciation

Find the value after depreciation.

- a) £15000 car, 20% depreciation, 3 years
- b) £800 laptop, 25% depreciation, 2 years
- c) £6000 motorbike, 10% depreciation, 4 years
- d) £20000 van, 12% depreciation, 5 years

4. Comparing simple and compound

Calculate both simple and compound interest, then find the difference.

- a) £1000 at 5% for 3 years: difference?
- b) £2000 at 4% for 4 years: difference?
- c) £5000 at 3% for 5 years: difference?
- d) £10000 at 6% for 3 years: difference?

5. Mixed problems

Solve each compound percentage problem.

- a) £3000 invested at 4% for 5 years. b) A phone worth £600 depreciates 20% each year. Value after 3 years? c) Population 50000 grows 2% per year. Population after 10 years? d) £4000 grows to £4410 in 2 years. Annual rate? e) House worth £200000 increases 6% per year for 3 years. New value?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Compound interest grows faster than simple interest ☐

Correct answer: _____

Reason: _____

b) 5% compound interest for 3 years is 15% total increase ☐

Correct answer: _____

Reason: _____

c) Depreciation of 20% per year uses multiplier 0.8 ☐

Correct answer: _____

Reason: _____

d) After 100% depreciation the item is worthless ☐

Correct answer: _____

Reason: _____

e) The compound interest formula is $A = P(1 + r)^n$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Priya invests **£5000** at **3.5% compound interest**. After how many complete years will her investment first exceed **£6000**? Show your working.

Expression: _____

Simplified: _____

b) A car loses **20%** of its value each year. It is currently worth **£24000**. How many years until it is worth less than **£10000**? What will its value be at that point?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

